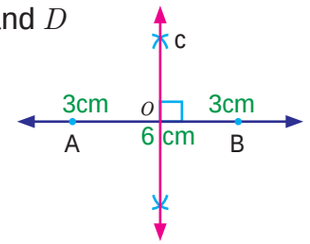
Step 3: With the same radius and B as center draw two arcs to cut the arcs drawn in step 2. Mark the points of intersection of the arcs as C and D

Step 4: Join C and D. CD will intersect AB. Mark the point of intersection as O

CD is the required perpendicular bisector of AB.

Measure $\angle AOC$. Measure the length of AO and OB. What do you observe?



Think

1. What will happen if the radius of the arc is less than half of AB?

Exercise 5.3

1. Draw a line segment of given length and construct a perpendicular bisector to each line segment using scale and compass.

   (a) 8 cm (b) 7cm (c) 5.6 cm (d) 10.4 cm (e) 58 mm

5.4.2 Construction of angle bisector of an angle.

If a line or line segment divides an angle into two equal angles, then the line or line segment is called angle bisector of the given angle.

In the following figures we can observe some angle bisectors.

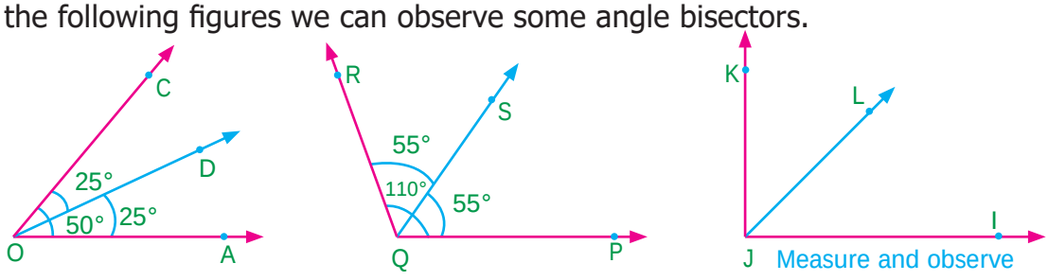


Fig.5.25

Example 5.12 Construct bisector of the $\angle ABC$ with the measure 80°.

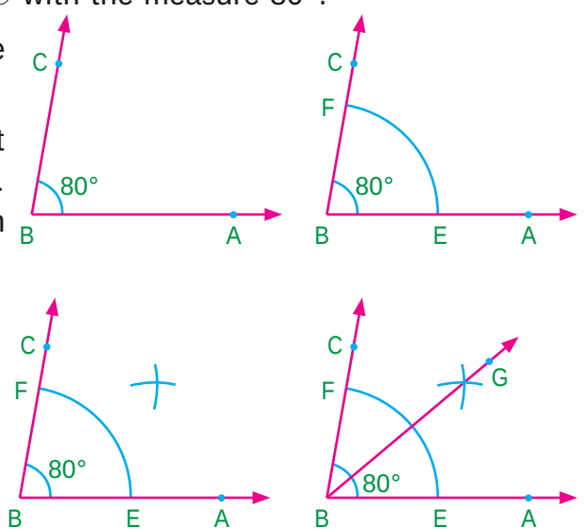
Step 1: Draw the given angle $\angle ABC$ with the measure 80° using protractor.

Step 2: With B as center and convenient radius, draw an arc to cut BA and BC. Mark the points of intersection as E on BA and F on BC.

Step 3: With the same radius and E as center, draw an arc in the interior of $\angle ABC$ and another arc of same measure with center at F to cut the previous arc.

Step 4: Mark the point of intersection as G. Draw a ray BX through G.

BG is the required bisector of the given angle $\angle ABC$



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